Bank of Portugal Course—Empirical Macroeconomics

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- **Overview.** This is a course for research staff at the Bank of Portugal and economics PhD students. We will study empirical macroeconomics, with a focus on both causality and identification, both using reduced form methods and structural estimation. Lectures will often take the form of studying a topic (e.g. the effect of monetary policy on the economy) and a method (e.g. local projections and VARs).
- **Organization.** The class will meet at the following dates and times:
  - Tuesday – October 28th:
    - 14h00 – 17h15 (with a 15-minute break)
  - Wednesday – October 29th:
    - 10h00 - 12h15 (with a 15-minute break)
    - 14h00 – 16h15(with a 15-minute break)
  - Thursday – October 30th:
    - 10h00 - 12h15 (with a 15-minute break)
    - 14h00 – 16h15 (with a 15-minute break)
  - Friday – October 31st:
    - 10h00 - 12h15 (with a 15-minute break)
    - 14h00 – 16h15 (with a 15-minute break)

Rough Plan

- Tuesday Lecture: Identification in Macro, Monetary and Fiscal Policy, and Hysteresis
  - Course overview
  - Identification in macro
  - Monetary policy
  - Fiscal policy
  - Hysteresis
- Wednesday Lectures: Phillips Curves and inflation
  - Time series evidence
  - Regional estimates
  - Inflation after the Pandemic
- Thursday Lectures: labor markets

  - New costs of inflation in the labor market
  - Wage rigidity and unemployment fluctuations

- Friday Lectures:
  - Evidence on models of consumption behavior
    - Time series evidence
    - Studies of marginal propensity to consume
    - Structural estimation of precautionary savings
    - Introduction to event study estimation
  - Automation
    - Effect of robots on labor market outcomes
    - Introduction to shift share instruments

References

Useful Advice on How to do Research

I am attaching various links about how to do research. I benefitted from all of these enormously during grad school.

- **Most important.** Matthias Doepke's guide to how to give a lunch talk.
- John Cochrane's guide to writing for PhD students.
- Jesse Shapiro's advice for writing an applied micro paper and how to give an applied micro talk.
- Adam Guren's advice about how to give a lunch talk.
- Eric Zwick's 12 step guide to grad school.
- Rachael Meager's guide to public speaking.
- David Romer's 5 step guide to grad school.

Tuesday Lecture: Identification in Macro, Monetary and Fiscal Policy, and Hysteresis

- Nakamura, Emi, and Jón Steinsson. "Identification in macroeconomics." Journal of Economic Perspectives 32.3 (2018): 59-86.
- Romer, C.D. and D.H. Romer (2004): "A New Measure of Monetary Shocks: Derivation and Implications," American Economic Review, 94(4), 1055-1084.
- Coibion, O. (2012): "Are the Effects of Monetary Policy Shocks Big or Small?" American Economic Journal: Macroeconomics, 4(2), 1-32.
- Ramey, Valerie A. "Macroeconomic shocks and their propagation." Handbook of macroeconomics 2 (2016): 71-162.
- Aruoba, S. Borağan, and Thomas Drechsel. "Identifying Monetary Policy Shocks Using Natural Language Processing." (2024).
- Nakamura, Emi, and Jón Steinsson. "High-frequency identification of monetary non-neutrality: the information effect." The Quarterly Journal of Economics 133.3 (2018): 1283-1330.

- Plagborg-Møller, Mikkel, and Christian K. Wolf. "Local projections and VARs estimate the same impulse responses." Econometrica 89.2 (2021): 955-980.
- Montiel Olea, José Luis, Plagborg-Møller, Mikkel, Qian, Eric and Wolf, Christian (2025): "Local Projections or VARs? A Primer for Macroeconomists." Working paper.
- Nakamura, E. and J. Steinsson (2014): "Fiscal Stimulus in a Monetary Union: Evidence from U.S. Regions," American Economic Review, 104(3), 753-792.
- Chodorow-Reich, Gabriel. "Geographic cross-sectional fiscal spending multipliers: What have we learned?." American Economic Journal: Economic Policy 11.2 (2019): 1-34.
- Ramey, Valerie A., and Sarah Zubairy. "Government spending multipliers in good times and in bad: evidence from US historical data." Journal of Political Economy 126.2 (2018): 850-901.
- Blanchard, Olivier, Eugenio Cerutti, and Lawrence Summers. "Inflation and activity–two explorations and their monetary policy implications. "No. w21726. National Bureau of Economic Research, 2015.

Wednesday Lectures: Inflation Before and After the Pandemic.

- **Required reading:** Hazell, J., Herreno, J., Nakamura, E., & Steinsson, J. (2020). The slope of the Phillips Curve: evidence from US states (No. w28005). National Bureau of Economic Research.
- Atkeson, Andrew E. Ohanian (2001). "Are Phillips Curves Useful for Forecasting Inflation?" Federal Reserve bank of Minneapolis quarterly review 25.1: 2-2.
- Ball, L., & Mazumder, S. (2014). A Phillips Curve with Anchored Expectations and Short-Term Unemployment. Journal of Money, Credit and Banking.
- Beraja, M., Hurst, E., & Ospina, J. (2019). "The aggregate implications of regional business cycles," Econometrica, 87(6), 1789-1833.
- Barnichon, Regis and Mesters, G. (2020). "Identifying Modern Macro Equations with Old Shocks," The Quarterly Journal of Economics, forthcoming.
- Gali, J. and M. Gertler (1999): "Inflation Dynamics: A Structural Econometrics Analysis," Journal of Monetary Economics, 44(2), 195-222.
- Mavroeidis, S., M. Plagborg-Moller, and James H. Stock (2014): "Empirical Evidence on Inflation Expectations in the New Keynesian Phillips Curve," Journal of Economic Literature, 52(1), 124-188.
- Beaudry, Paul, Chenyu Hou, and Franck Portier. On the Fragility of the Nonlinear Phillips Curve View of Recent Inflation. No. w33522. National Bureau of Economic Research, 2025.
- Hazell, Jonathon, and Stephan Hobler. Do Deficits Cause Inflation?: A High Frequency Narrative Approach. London School of Economics and Political Science, 2024.